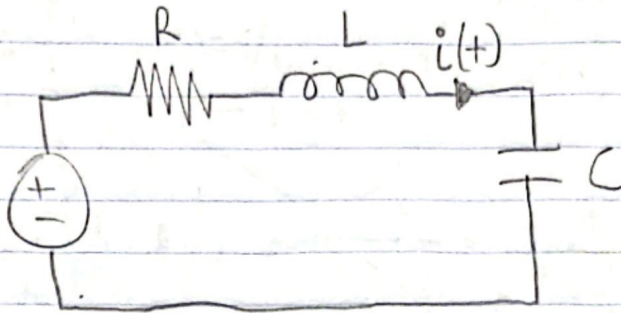


$$1/c \int i dt$$

$$L \frac{di}{dt}$$

$$Ri$$

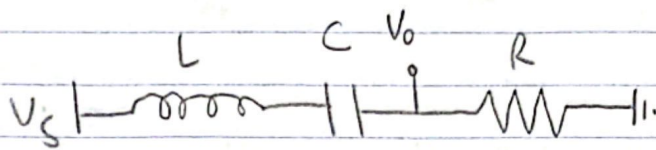
3)



$$Z_{Z_1} + Z_{Z_2} + Z_{Z_1} + Z_{Z_2}$$

$$(Z_{Z_1} + Z_{Z_2})^2$$

a)



$$Z_T = Z_L + Z_C + Z_R$$

$$= Ls + 1/sC + R$$

$$V_o = R \cdot i_s(t)$$

$$H(s) = \frac{R}{sL + 1/sC + R} \cdot \frac{s/L}{s/L} = \frac{\frac{RS}{L}}{s^2 + \frac{RS}{L}s + \frac{1}{LC}} = \frac{KS}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{V_o}{V_{in}}$$

$$V_o = V_i \cdot \frac{\frac{RS}{L}}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \rightarrow \alpha = 2\frac{R}{L}, \omega_0 = \frac{1}{\sqrt{LC}}$$

b)

$$10^6 = \frac{1}{\sqrt{(10^{-6})(C)}} \rightarrow \sqrt{10^{-6}C} = \frac{1}{10^6} \rightarrow 10^{-6}C = \frac{1}{10^{12}}$$

$$C = \frac{1}{10^{12}} = 1 \mu F$$

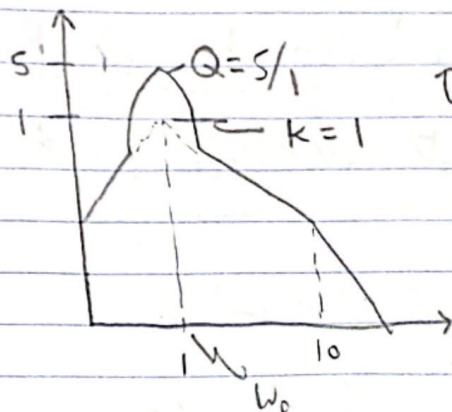
$$R = 10^3$$

$$c) \quad \frac{\frac{R s}{L}}{s^2 + \frac{R s}{L} + 1/LC} \quad \left| \begin{array}{l} L = C = 10^{-6} \\ s = j10^6 \end{array} \right.$$

$$I = \frac{\frac{R s}{10^{-6}}}{s^2 + \frac{R s}{10^{-6}} + 1/10^{-12}} = \frac{\frac{jR10^6}{10^{-6}}}{\underbrace{j\cancel{\omega_0^2} + \frac{jR10^{-6}}{10^{-6}}}_{- \omega_0^2} + \underbrace{1/LC}_{\omega_0^2}} = \frac{jR}{jR} = 1 \quad | \quad R \text{ doesn't affect mag!}$$



4)



$$\tau = 1/10 = 0.15$$

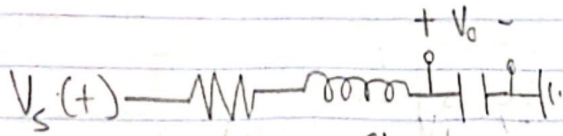
5)

a) Cannot oscillate with only RC, needs L.  
False

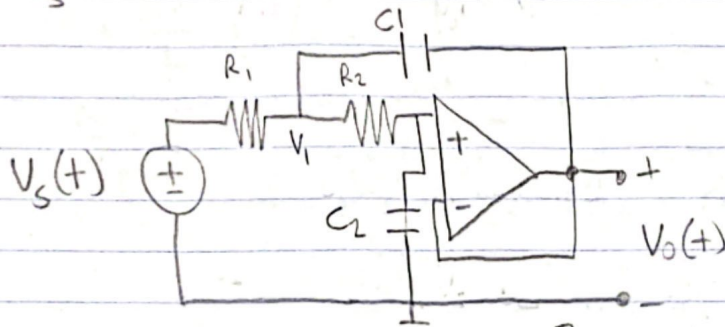
b)  $w_0 = 1/\sqrt{LC}$   
False

c) Input source  $f = w_0$ ,  $Z_C \neq Z_L = 0$ , short to ground,  $V_0 = 0$ .  
True.

6 a)



b)



$$\text{For a: } H(s) = \frac{V_{out}}{V_{in}} = \frac{Z_C}{Z_R + Z_L + Z_C} = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} \cdot \frac{sC}{sC}$$

$$= \frac{1}{sRC + s^2LC + 1} = \frac{1/LC}{s^2 + \frac{R}{L}s + 1/LC}$$

$$\text{For b: } H(s) = \frac{V_{out}}{V_{in}}$$

$$V_{out} = V^- = V^+$$

$$V^+: \frac{V^+ - 0}{1/sC_2} + \frac{V^+ - V_1}{R_2} = V^+ sC_2 + \frac{V^+ - V_1}{R_2} = 0$$

$$V_1: \frac{V_1 - V_s}{R_1} + \frac{V_1 - V^+}{R_2} + \frac{V_1 - V_o}{1/sC_1} = \frac{V_1 - V_s}{R_1} + \frac{V_1 - V^+}{R_2} + (V_1 - V_o)sC_1$$

$$\text{Eq: } V^+: V_o sC_2 + (V_o - V_1)/R_2 = 0$$

$$V_1: (V_1 - V_s)/R_1 + (V_1 - V_o)/R_2 + (V_1 - V_o)sC_1$$



$$6) V_0(sC_2) + (V_0 - V_1)/R_2 = 0 \Rightarrow V_0(sR_2C_2) + V_0 = V_1$$

$$\boxed{R_2 V_0 / (sC_2) + V_0 = V_1} \quad V_1 = V_0(sR_2C_2 + 1)$$

$$(V_1 - V_s)/R_1 + (V_1 - V_0)/R_2 + (V_1 - V_0)sC_1 = 0$$

$$\Rightarrow \left[ \frac{(R_2 V_0 / (sC_2) + V_0) - V_s}{R_1} + \frac{(R_2 V_0 / (sC_2) + V_0) - V_0}{R_2} + \frac{(R_2 V_0 / (sC_2) + V_0) - V_0}{sC_1} \right]$$

$$\Rightarrow \left[ \frac{(V_0(sR_2C_2 + 1) - V_s)}{R_1} + \frac{(V_0(sR_2C_2 + 1) - V_0)}{R_2} + \frac{(V_0(sR_2C_2 + 1) - V_0)}{sC_1} \right]$$

$$\Rightarrow \frac{V_0(sR_2C_2 + 1) - V_s}{R_1} + V_0C_2s + V_0R_2C_2C_1s^2$$

$$\Rightarrow V_0(sR_2C_2 + 1) - V_s + sV_0R_1C_2 + s^2V_0R_1R_2C_1C_2 = 0$$

$$V_s = V_0(sR_2C_2 + 1) + sV_0R_1C_2 + s^2R_1R_2C_1C_2V_0$$

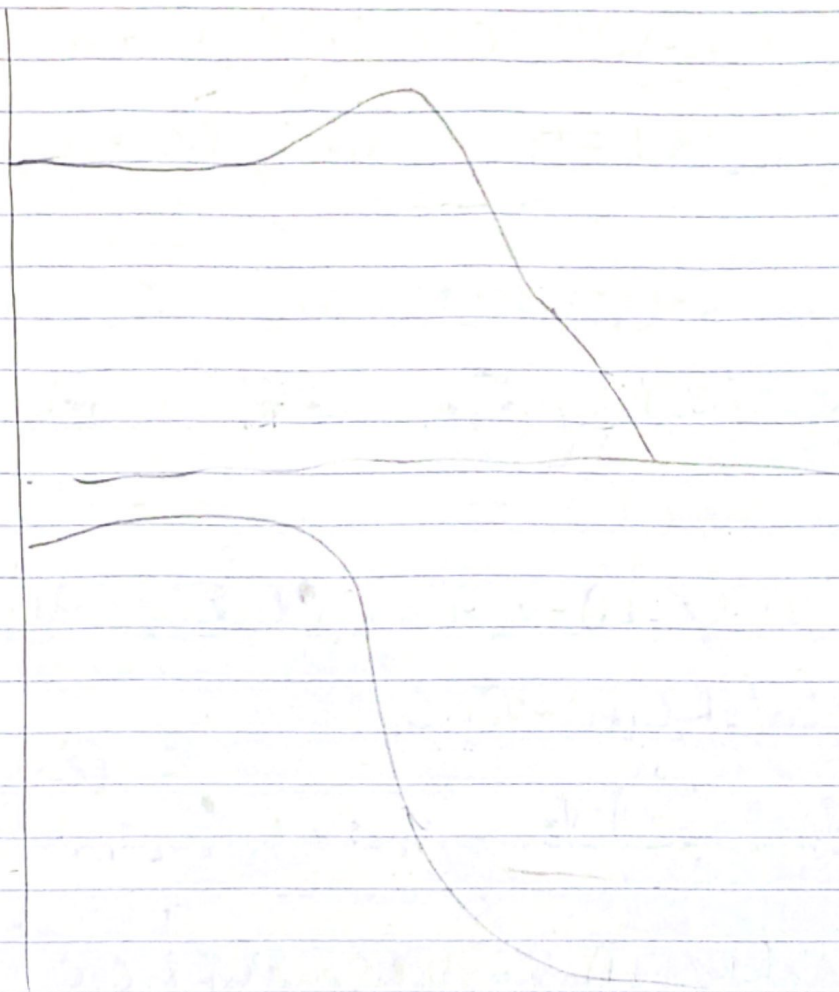
$$\frac{V_0}{V_s} = \frac{1}{sR_2C_2 + 1 + sR_1C_2 + s^2R_1R_2C_1C_2}$$

$$= \frac{1}{s^2R_1R_2C_1C_2 + s(R_1 + R_2)C_2 + 1}$$

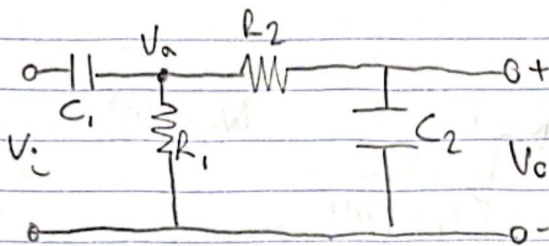
Same form!  $as^2 + bs + 1$

MAG

Phase







Nodal:

$$V_a: (V_a - V_i) s C_1 + V_a / R_1 + \frac{V_a - V_o}{R_2} = 0$$

$$V_o: \frac{V_o - V_a}{R_2} + V_o s C_2 = 0$$

$$\Rightarrow \frac{V_o}{R_2} - \frac{V_a}{R_2} + V_o s C_2 = 0$$

$$\Rightarrow V_o \left( \frac{1}{R_2} + s C_2 \right) - \frac{V_a}{R_2} = 0$$

$$V_o = \frac{V_a / R_2}{\left( \frac{1}{R_2} + s C_2 \right)} = \frac{V_a}{1 + s R_2 C_2} = \frac{V_a}{1 + s}$$

Plugging in:  $R_1 = R_2 = 1k$ ,  $C_1 = C_2 = 1\mu F$ :

$$V_o = \frac{V_a}{1 + s}, \quad V_a \text{ node: } \frac{(V_a - V_i) s}{1000} + \frac{V_a}{1000} + \frac{V_a - V_o}{1000} = 0$$

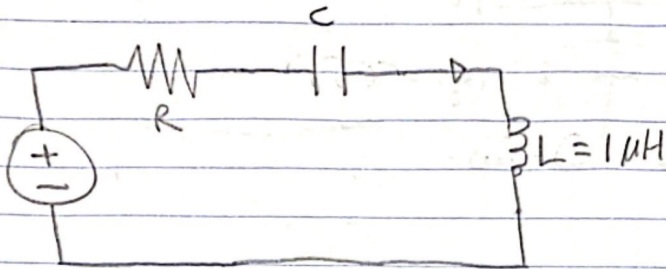
$$\Rightarrow \frac{1}{1000} ((V_a - V_i) s + V_a + V_a - V_o) = 0$$

$$V_a (s + 2) - s V_i - V_o = 0 \Rightarrow V_a = \frac{s V_i + V_o}{s + 2} = V_o \frac{s}{s + 2}$$

$$s V_i + V_o = V_o (1 + s)(s + 2)$$

$$s V_i = V_o (s^2 + 3s + 2 - 1) \Rightarrow \frac{V_o}{V_i} = \frac{s}{s^2 + 3s + 1}$$

3)



$$\omega_0 = 10^6$$

$$V_o = \frac{Z_L}{Z_C + Z_L + Z_R} \cdot V_i = \frac{LS}{1/sC + Ls + R} = V_i \left( \frac{s^2 LC}{1 + s^2 LC + sRC} \right)$$

$$= V_i \left( \frac{s^2}{s^2 + sR/L + 1/LC} \right) \quad \alpha = R/L \quad \omega_0 = 1/\sqrt{LC}$$

$$\omega_0^2 = \frac{1}{LC} \rightarrow C = \frac{1}{L \cdot \omega_0^2} = \frac{1}{10^6} = 1 \mu F$$

$$1 = \frac{-\omega_0^2}{-\omega_0^2 + j\omega_0 \frac{R}{L} + 1/LC} = \frac{-1/LC}{-1/LC + j\omega_0 \frac{R}{L} + 1/LC} = \frac{-1/LC}{j\omega_0 R/L} = \frac{-\omega_0}{jR/L}$$

$$1 = \left| \frac{-\omega_0}{j(R/L)} \right| = \frac{\omega_0}{R/L}$$

$$\frac{R}{L} = \omega_0 \rightarrow R = \omega_0(L) = 1 \Omega$$



7)

